

Hardy Space Formalization Blueprint

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Chapter 1

Canonical factorization in the Nevanlinna class

Project overview. This project develops a Lean 4 and mathlib formalization of Hardy spaces and the Nevanlinna class on the unit disc. Its ultimate goal is a formal proof of the canonical factorization theorem for the Nevanlinna class, decomposing a nonzero function into its Blaschke, singular, and outer factors. This blueprint records the mathematical dependency graph and the current formalization status.

Main reference. The main mathematical reference is John B. Garnett, *Bounded Analytic Functions*. Unless another source is stated, the theorem numbers and chapter/section references below refer to Garnett’s book. The statements in this blueprint are formalization-oriented paraphrases and are not verbatim reproductions of the source.

1.1 Chapter I prerequisites

Theorem 1 (Harmonic majorants of subharmonic functions, Garnett I.6.7). *Subharmonic.hasHarmonicMajorant iff, Garnett, Chapter I, Section 6, PDF page 36. Let v be subharmonic on D . Then v has a harmonic majorant on D if and only if $\sup_{0 < r < 1} v_r(0) < \infty$, where v_r is the Poisson extension to $|z| < r$ of the boundary values of v on $|z| = r$. When this condition holds, the least harmonic majorant u of v is given by*

$$u(z) = \lim_{r \rightarrow 1} \frac{1}{2\pi} \int_0^{2\pi} P_{z/r}(\theta) v(re^{i\theta}) d\theta = \lim_{r \rightarrow 1} v_r(z).$$

1.2 Nontangential limits and boundary values

Definition 2 (Nontangential approach region). Source: Garnett, Chapter I, Section 4, PDF page 23 (disc form). For $\zeta \in \partial\mathbb{D}$ and $\alpha > 1$ put

$$\Gamma_\alpha(\zeta) = \left\{ z \in \mathbb{D} : |\zeta - z| < \alpha(1 - |z|) \right\},$$

a teardrop-shaped region asymptotic at ζ to an angle with vertex ζ . Facts to record: $\Gamma_\alpha(\zeta)$ is open and contained in $\mathbb{D}$; it increases with α ; it contains the radial segment $[0, 1)\zeta$; it is empty when $\alpha \leq 1$; and $\Gamma_\alpha(1)$ is mathlib’s `Complex.stolzSet  $\alpha$` , with $\Gamma_\alpha(\zeta) = \zeta \cdot \Gamma_\alpha(1)$. Mathlib only has the region at $\zeta = 1$, so the rotated family has to be added.

Definition 3 (Nontangential limit). *def:nt_regionSource : Garnett, Chapter I, Section 5, PDF page 27. Write* $N_{\Gamma_\alpha(\zeta)}(\zeta)$ *for the neighbourhood filter of* ζ *within* $\Gamma_\alpha(\zeta)$. The *nontangential filter* at ζ is their supremum in the lattice of filters,

$$\mathcal{N}_\zeta = \sup_{\alpha > 1} \mathcal{N}_{\Gamma_\alpha(\zeta)}(\zeta).$$

For $\alpha > 1$, f has limit L at ζ *within* $\Gamma_\alpha(\zeta)$ if $f \rightarrow L$ along $\mathcal{N}_{\Gamma_\alpha(\zeta)}(\zeta)$, and f has *nontangential limit* L at ζ if $f \rightarrow L$ along $\mathcal{N}_\zeta$. The two are related by

$$f \rightarrow L \text{ along } \mathcal{N}_\zeta \iff \forall \alpha > 1, f \rightarrow L \text{ along } \mathcal{N}_{\Gamma_\alpha(\zeta)}(\zeta).$$

Definition 4 (Boundary value). *def:nt_limitFor* $\alpha > 1$ the boundary value of f at ζ within a single approach region, and the boundary value of f at ζ , are

$$f_\alpha^*(\zeta) = \lim_{\Gamma_\alpha(\zeta) \ni z \rightarrow \zeta} f(z), \quad f^*(\zeta) = \lim_{\mathcal{N}_\zeta} f,$$

the limits along $\mathcal{N}_{\Gamma_\alpha(\zeta)}(\zeta)$ and along the nontangential filter $\mathcal{N}_\zeta$, read through the choice operator so that both are total functions. Where the nontangential limit at ζ exists, it equals $f^*(\zeta)$, and $f_\alpha^*(\zeta) = f^*(\zeta)$ for every $\alpha > 1$.

Lemma 5 (Poisson kernel over a Stolz region). *poissonKernel_e_of_m_stolz_Set_At_def : nt_region.Let* $\alpha > 1$. *There is a constant* C_α *such that for every* $\zeta \in \partial\mathbb{D}$, *every* $z \in \Gamma_\alpha(\zeta)$ *and every* θ ,

$$P_z(\theta) \leq \frac{C_\alpha (1 - |z|)}{((1 - |z|) + |e^{i\theta} - \zeta|)^2}.$$

In particular $P_z(\theta) \leq C_\alpha/(1 - |z|)$ *always, and* $P_z(\theta) \leq C_\alpha(1 - |z|)/|e^{i\theta} - \zeta|^2$ *once* $|e^{i\theta} - \zeta| \geq 2\alpha(1 - |z|)$. *Together with* $\int P_z d\theta/2\pi = 1$ *this exhibits the Poisson kernels, restricted to a Stolz region, as an approximate identity with a uniformly integrable decreasing majorant. This single estimate replaces Garnett's Theorem I.4.2 and is the engine of both Lemma 7 and Theorem 8.*

Proof. *lem:poisson_stolz_boundFrom* $|e^{i\theta} - z| \geq 1 - |z|$ and, by the triangle inequality and the definition of $\Gamma_\alpha(\zeta)$, $|e^{i\theta} - z| \geq |e^{i\theta} - \zeta| - \alpha(1 - |z|)$, one gets $|e^{i\theta} - z| \geq c_\alpha((1 - |z|) + |e^{i\theta} - \zeta|)$; insert this into $P_z(\theta) = (1 - |z|^2)/|e^{i\theta} - z|^2$ and use $1 - |z|^2 \leq 2(1 - |z|)$. $\square$

Theorem 6 (Differentiation of measures on the circle). *Let* ν *be a finite positive Borel measure on* $\partial\mathbb{D}$ *and let* $I(\zeta, h)$ *denote the arc of length* $2h$ *centred at* ζ . *Then for almost every* ζ ,

$$\lim_{h \rightarrow 0} \frac{\nu(I(\zeta, h))}{2h} = \frac{d\nu}{d\theta}(\zeta),$$

and in particular $\nu(I(\zeta, h))/2h \rightarrow 0$ *almost everywhere when* $\nu \perp d\theta$. *In the same setting, almost every* ζ *is a Lebesgue point of a given* $k \in L^1(d\theta/2\pi)$:

$$\lim_{h \rightarrow 0} \frac{1}{\mu(I(\zeta, h))} \int_{I(\zeta, h)} |k - k(\zeta)| d\mu = 0, \quad \mu = d\theta/2\pi.$$

Both halves come from *mathlib* *applied directly on* $\mathbb{C}$ *with* $\mu = \text{circleMeasure } 0 \ 1$, *with no passage to intervals in* θ : *Besicovitch.ae_tendsto_rnDeriv* *needs only local finiteness of the two measures,* *Besicovitch.vitaliFamily* *needs only* s -*finiteness, and* *VitaliFamily.ae_tendsto_average_norm_sub* *then gives the Lebesgue points;* *HasBesicovitchCovering* *holds for every finite-dimensional real normed space, in particular for* $\mathbb{C}$. *The singular case follows since the Radon–Nikodym derivative of a mutually singular measure vanishes almost everywhere.*

Two points of care. The Vitali families of `MeasureTheory.Covering.DensityTheorem` are not usable here: they require `IsUnifLocDoublingMeasure`, which fails for `circleMeasure` viewed as a measure on $\mathbb{C}$, since balls centred slightly off the circle have measure zero while their doubles do not; the Besicovitch family carries no such hypothesis. And $B(\zeta, h) \cap \partial\mathbb{D}$ is the arc $I(\zeta, h)$, so the only genuine work is the comparability $\mu(B(\zeta, h)) \asymp h$ of chord and arc, which Lemma 5 needs anyway. This node replaces Garnett's Vitali covering lemma I.4.4 and the weak-type argument built on it.

Lemma 7 (Garnett I.5.4: singular Poisson integrals vanish at the boundary). *def:ntlimit, lem : poisson_s_tolz_bound, thm : circle_differentiation* Source : Garnett, Chapter I, Section 5, PDF pages 41–42. If ν is a finite signed measure on $\partial\mathbb{D}$ that is singular with respect to $d\theta$, then

$$\lim_{z \rightarrow \zeta} P[\nu](z) = 0$$

for almost every $\zeta \in \partial\mathbb{D}$. This is the step that removes the singular part in Theorem 10, and it is also what gives the singular functions of Definition 29 boundary modulus one almost everywhere.

Proof. By the Jordan decomposition we may assume $\nu \geq 0$. Fix ζ at which $\nu(I(\zeta, h))/2h \rightarrow 0$, which by Theorem 6 is almost every ζ . For $z \in \Gamma_\alpha(\zeta)$ split the integral at $|e^{i\theta} - \zeta| = A(1 - |z|)$. On the far part Lemma 5 gives a bound $C_\alpha \nu(\partial\mathbb{D})/(A^2(1 - |z|))$, and on the near part the same lemma bounds the integral by $C_\alpha \sup_{h < A(1 - |z|)} \nu(I(\zeta, h))/2h$. Choose $A = A(z)$ with $A(1 - |z|) \rightarrow 0$ and $A^2(1 - |z|) \rightarrow \infty$ as $|z| \rightarrow 1$. $\square$

Theorem 8 (Fatou at Lebesgue points). *def:ntlimit, lem : poisson_s_tolz_bound, thm : circle_differentiation* Let $k \in L^1(d\theta/2\pi)$. Then at every Lebesgue point ζ of k , and hence almost everywhere,

$$\lim_{z \rightarrow \zeta} P[k d\theta/2\pi](z) = k(\zeta).$$

Proof. Estimate $|P[k](z) - k(\zeta)| \leq \int P_z(\theta) |k(\theta) - k(\zeta)| d\theta/2\pi$ using $\int P_z d\theta/2\pi = 1$, and split the integral into dyadic arcs around ζ ; the majorant of Lemma 5 converts the Lebesgue-point condition into a convergent sum. This is the standard approximate-identity argument and it avoids the maximal function entirely. $\square$

Theorem 9 (Lebesgue decomposition of a finite boundary measure). A finite signed Borel measure μ on $\partial\mathbb{D}$ decomposes uniquely as

$$d\mu = k \frac{d\theta}{2\pi} + d\mu_s, \quad k \in L^1(d\theta/2\pi), \quad \mu_s \perp d\theta.$$

This is available in `mathlib: MeasureTheory.SignedMeasure.singularPart_add_withDensity_rnDeriv_eq, SignedMeasure.rnDeriv, SignedMeasure.integrable_rnDeriv`, the signed density operator `Measure.withDensity` for vector measures, and the mutual-singularity relation `VectorMeasure.MutuallySingular`. Only the ambient space has to be fixed, which the convention at the head of this section does.

Theorem 10 (Garnett I.5.3: Fatou theorem, $p = 1$ disc form). *thm: I53a, lem : I54, thm : signed_lebesgue_decomposition, def : ntlimit, def : boundary_function* Source : Garnett, Chapter I, Section 5, PDF page 41. Let μ be a finite signed measure on $\partial\mathbb{D}$, with Lebesgue decomposition $d\mu = k d\theta/2\pi + d\mu_s$, $\mu_s \perp d\theta$. Then the Poisson integral of μ has a nontangential limit at almost every boundary point, and

$$\lim_{z \rightarrow \zeta} P[\mu](z) = k(\zeta) \quad \text{for almost every } \zeta \in \partial\mathbb{D};$$

equivalently $(P[\mu])^* = k$ almost everywhere, so the boundary function is in L^1 . Garnett states the stronger half-plane theorem for all $1 \leq p \leq \infty$; only this $p = 1$ disc form is used in this project.

Proof. Split μ by Theorem 9 and add Theorem 8 to Lemma 7, using that nontangential limits are limits along a filter and therefore additive. $\square$

Theorem 11 (Bounded harmonic functions are Poisson integrals of L^∞ data). *thm:positive_harmonic_poisson_measure* Let M , then there is $k \in L^\infty(d\theta/2\pi)$ with $\|k\|_\infty \leq M$ and $u = P[k d\theta/2\pi]$.

Proof. Apply Theorem 24 to the positive harmonic function $u + M$ to get $u + M = P[\mu]$ with $\mu \geq 0$ finite. Since μ is the weak-star limit of the measures $(u(r \cdot) + M) d\theta/2\pi$, whose densities are bounded by $2M$, one gets $\mu \leq 2M d\theta/2\pi$; Radon–Nikodym then produces a density bounded by $2M$, and subtracting M gives k . $\square$

Theorem 12 (A boundary measure is determined by its Poisson integral). *If μ and ν are finite signed measures on $\partial\mathbb{D}$ with $P[\mu] = P[\nu]$ on $\mathbb{D}$, then $\mu = \nu$. The proof is the standard weak-star argument: $P[\mu](r \cdot) d\theta/2\pi \rightarrow \mu$ when tested against continuous functions on the circle. This is what licenses speaking of the representing measure in Theorem 27, and it is used again for the uniqueness of the singular factors (Theorem 33) and in Garnett’s Theorem II.5.4.*

1.3 Chapter II, Section 2: Blaschke products

Theorem 13 (Garnett II.2.1: harmonic majorants imply the Blaschke condition). *summable_onesubnorm_of_harmonic* Garnett, Chapter II, Section 2, PDF page 63. Let f be a non-zero analytic function on D , and let $\{a_n\}$ be its zeros, counted with multiplicity. If $\log |f|$ has a harmonic majorant on $\mathbb{D}$, then

$$\sum_n (1 - |a_n|) < \infty.$$

Moreover, if $f(0) \neq 0$ and u is the least harmonic majorant of $\log |f|$, then

$$\sum_n (1 - |a_n|) \leq u(0) - \log |f(0)|.$$

By Theorem 22, the harmonic-majorant hypothesis is equivalent to f belonging to the Nevanlinna class N .

Proof. After removing a possible zero at the origin, fix $r < 1$ avoiding the moduli of the zeros and divide $f(rz)$ by the finite Blaschke product with the same zeros. The logarithm of the modulus of the resulting zero-free analytic function is harmonic, so the mean-value formula gives the familiar Jensen identity

$$\log |f(0)| + \sum_{|a_j| < r} \log \frac{r}{|a_j|} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta.$$

Theorem 1 identifies the limit of the right-hand side with $u(0)$. Letting $r \rightarrow 1$ and using $1 - |a_j| \leq \log(1/|a_j|)$ proves both conclusions. $\square$

Theorem 14 (Garnett II.2.2: convergence and boundary size of Blaschke products). *Source: Garnett, Chapter II, Section 2, PDF page 64.* If $\{a_n\} \subset \mathbb{D}$ satisfies the Blaschke condition, then the associated Blaschke product converges uniformly on compact subsets of $\mathbb{D}$ to a bounded analytic function whose zeros are exactly the a_n with the prescribed multiplicities, and whose boundary modulus is one almost everywhere.

Theorem 15 (Garnett II.2.3 (F. Riesz): division by the Blaschke factor). *thm:II₂₁, thm : II₂₂Source : Garnett, Chapter II, Section 2, PDF page 66. Let $0 < p < \infty$, let $f \in H^p(\mathbb{D})$ be nonzero, and let B be the Blaschke product formed from all zeros of f , counted with multiplicity. Then $g = f/B$ extends analytically across the removable singularities, belongs to $H^p(\mathbb{D})$, and satisfies*

$$\|g\|_{H^p} = \|f\|_{H^p}.$$

Theorem 16 (Garnett II.2.4: characterization of Blaschke products). *thm:I₆₇, thm : II₂₁, thm : II₂₂Source : Garnett, Chapter II, Section 2, PDF page 66. Let $f \in H^\infty(\mathbb{D})$ with $\|f\|_\infty \leq 1$. Then the following are equivalent:*

1. $f = \lambda B$, where $|\lambda| = 1$ and B is a Blaschke product;

2.

$$\lim_{r \rightarrow 1} \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta = 0;$$

3. the least harmonic majorant of $\log |f|$ is the zero function.

The equivalence of the last two conditions is the direct use of Theorem 1; condition (3) is the form used in Lemma 25.

Definition 17 (Lean Blaschke factors). *BlaschkeFactor, normedBlaschkeFactor thm:II₂₁, thm : II₂₂, thm : II₂₃, thm : II₂₄The current Lean project has the one-zero Blaschke factors. The later formalization targets the modulus theorem, prove the H^p quotient theorem II.2.3, and provide the Blaschke-product characterization II.2.4 used for Nevanlinna factorization.*

1.4 Chapter II, Section 4: boundary uniqueness and outer functions

Corollary 18 (Garnett II.4.2: boundary uniqueness). *thm:II₅₃, def : boundary_functionSource : Garnett, Chapter II, Section 4, PDF page 75. Let $f \in N$, and suppose the boundary function f^* vanishes on a subset of $\partial\mathbb{D}$ of positive Lebesgue measure. Then $f \equiv 0$. Since $H^p \subseteq N$ for every $p > 0$, this contains Garnett's statement for $H^p(\mathbb{D})$, $0 < p \leq \infty$.*

Proof. Contrapositive. If $f \not\equiv 0$ then part (2) of Theorem 27 gives $\log |f^*| \in L^1(d\theta/2\pi)$, so $\log |f^*| > -\infty$ and hence $f^* \neq 0$ almost everywhere. The passage from H^p to N is the inclusion `hpDisc_le_nevanlinnaDisc`, already formalized.

Garnett argues the other way round, deducing this from Theorem II.4.1, whose proof dominates $\log^+ |f(re^{i\theta})|$ by $(1/p)|f^*|^p$ with f^* the nontangential maximal function of Theorem II.3.1. Reversing the implication keeps the present development free of the maximal function and of the Hardy–Littlewood theory behind it, at the cost of making this corollary depend on Theorem 27. There is no circularity: Theorem 27 obtains the nonvanishing of the boundary values of $e^{-(u_j + iv_j)}$ from a second application of Theorem 10, not from this corollary. What is lost is Garnett's sharper inequality II.4.1 itself, $\log |f(z_0)| \leq \int \log |f^*| P_{z_0} d\theta/2\pi$ for $f \in H^p$, which is not needed for Theorem 35; it would be needed for the N^+ refinements, and there the maximal function would have to be developed anyway. $\square$

Theorem 19 (Garnett II.4.4: construction from logarithmic data). *Source: Garnett, Chapter II, Section 4, PDF page 75. Given a nonnegative boundary function φ with $\log \varphi \in L^1(\partial\mathbb{D})$, there is an outer analytic function F on $\mathbb{D}$ whose a.e. boundary modulus is φ . Equivalently, F is obtained by exponentiating the analytic function whose real part is the Poisson integral of $\log \varphi$.*

$\|f(z)\| \leq \|f\|_\infty$ pointwise on $\mathbb{D}$, and the bound passes to the limit along the filter. Garnett's Corollary II.3.2 itself is the characterization of which L^p functions are boundary values of H^p functions; that characterization and the remaining H^p cases are not needed here. $\square$

Theorem 27 (Garnett II.5.3: boundary values and the least-majorant measure). *thm:II₅₁, lem : II₅₂, thm : II₂₂, thm : I₅₃, cor : II₃₂, thm : signed_lebesgue_decomposition, thm : poisson_measure_unique, def : ntlimit, def : boundary_function* Source : Garnett, Chapter II, Section 5, PDF page 79. Let $f \in N$, $f \not\equiv 0$. Then:

1. f has a nontangential limit $f^*(\zeta)$ at almost every $\zeta \in \partial\mathbb{D}$, and $f^* \neq 0$ almost everywhere;
2. $\log |f^*| \in L^1(d\theta/2\pi)$;
3. the least harmonic majorant of $\log |f|$ is $P[\mu]$ for a unique finite signed measure μ on $\partial\mathbb{D}$, and

$$d\mu = \log |f^*| \frac{d\theta}{2\pi} + d\mu_s, \quad \mu_s \perp d\theta. \quad (1.1)$$

In other words, the absolutely continuous part of the representing measure is exactly the boundary logarithm, and the whole obstruction to the inequality $\log |f(z)| \leq \int \log |f^*| P_z d\theta/2\pi$ is concentrated in the singular part; Garnett's class N^+ is the subclass where $\mu_s \leq 0$.

Proof. Let B be the Blaschke product over the zeros of f and $g = f/B$. By Theorem 22 and Lemma 25, g is zero-free and $\log |g| = P[\mu]$ is the least harmonic majorant of $\log |f|$; by Theorem 12 the measure μ is unique. Write $d\mu = k d\theta/2\pi + d\mu_s$ using Theorem 9. By Theorem 10, $\log |g|$ has nontangential limit k almost everywhere, and since $|B^*| = 1$ almost everywhere (Theorem 14) the same holds for $\log |f|$. So $k \in L^1$ and (1.1) follow once f itself is known to have nontangential limits.

For that, decompose $\log |g| = u_1 - u_2$ with $u_j = P[\mu_j] \geq 0$ harmonic, $\mu = \mu_1 - \mu_2$ the Jordan decomposition. Choose harmonic conjugates v_j with $v_j(0) = 0$, which exist because the disc is simply connected and which mathlib provides as `InnerProductSpace.HarmonicOnNhd.exists_analyticOnNhd_ball`. then $\log g = (u_1 + iv_1) - (u_2 + iv_2) + ic$ for a real constant c , so that

$$g = e^{ic} \frac{e^{-(u_2 + iv_2)}}{e^{-(u_1 + iv_1)}}.$$

Each $e^{-(u_j + iv_j)}$ has modulus $e^{-u_j} \leq 1$, hence lies in H^∞ and has nontangential limits almost everywhere by Corollary 26. Those limits are nonzero almost everywhere: $\log |e^{-(u_j + iv_j)}| = -P[\mu_j]$ has a finite nontangential limit almost everywhere by Theorem 10 again, so the boundary modulus is $e^{-k_j} > 0$ almost everywhere. Hence g , and then $f = Bg$, have nontangential limits almost everywhere.

Garnett obtains the nonvanishing from the boundary uniqueness theorem (Corollary 18) instead. Using Theorem 10 as above keeps II.5.3 independent of the H^p boundary uniqueness theory, which is why `cor:II.4.2` is not among the dependencies above; it is still needed for the outer factor. $\square$

1.6 Singular factors

Theorem 28 (Jordan decomposition of singular measures). *If ν is a finite real signed measure on $\partial\mathbb{D}$ and is singular with respect to Lebesgue measure, then its Jordan decomposition $\nu = \nu^+ - \nu^-$ has ν^+ and ν^- finite positive singular measures.*

Definition 29 (Singular function formula). For a finite positive singular measure σ on $\partial\mathbb{D}$, the associated singular function is the analytic exponential

$$S_\sigma(z) = \exp \left(- \int_{\partial\mathbb{D}} \frac{\zeta + z}{\zeta - z} d\sigma(\zeta) \right).$$

It is bounded by one in the disc and has boundary modulus one almost everywhere.

Definition 30 (Singular factors). *thm:jordan_decomposition_singular_measure, def : singular_function_formula, lem : I_54* If the singular part of the representing measure is $\nu_s = \nu_s^+ - \nu_s^-$, then the canonical factorization uses the two singular functions $S_1 = S_{\nu_s^+}$ and $S_2 = S_{\nu_s^-}$.

1.7 Final factorization and uniqueness

Theorem 31 (Zeros determine the Blaschke factor). *The multiset of zeros in $\mathbb{D}$ determines the associated Blaschke product up to the chosen unimodular normalization. With a fixed convention for elementary factors, the product is uniquely determined.*

Theorem 32 (Boundary modulus determines the outer factor). *thm:outer_uniqueness* If two outer functions have the same

Theorem 33 (Singular measure determines singular factors). *def:singular_factors* The positive and negative singular parts ν_s^+ and ν_s^- of a singular measure ν_s determine the singular functions S_1 and S_2 . Conversely, equality of the corresponding singular functions forces equality of the singular measures, modulo the normalization convention.

Theorem 34 (Uniqueness of Nevanlinna factorization). *thm:zeros_determine_blaschke, thm : boundary_modulus_determines_outer, thm : singular_measure_determines_factors* In the canonical factorization, the Blaschke product B , the outer function F , and the singular functions S_1, S_2 are uniquely determined.

Theorem 35 (Garnett II.5.5: canonical factorization in N). *thm:II_51, lem : II_52, thm : II_53, def : outer_factor, def : singular_factors, thm : nevanlinna_factorization_uniqueness* Source : Garnett, Chapter II, Section 5.5. If f is in the Nevanlinna class N and f is not identically zero, then there exist a unimodular constant C , a Blaschke product B , an outer function F , and singular functions S_1, S_2 such that

$$f = C B F S_1 / S_2.$$

Conversely, every expression of this form belongs to the Nevanlinna class. The factors are unique after fixing the standard normalizations.

Proof. Remove the zeros using the Blaschke product. Apply the boundary-value and logarithmic-integrability theorem to the zero-free quotient. Split the representing measure of the least harmonic majorant into absolutely continuous and singular parts. The absolutely continuous part gives the outer factor, while the positive and negative singular parts give S_1 and S_2 . The uniqueness statement follows by comparing zero sets, boundary moduli, and singular measures. $\square$